

Executive Summary

Incremental “marginal gains” across propulsion, structures, avionics and operations can collectively boost a NASA-class deep-space rocket’s performance by tens of percent. Each domain (engines, tanks, guidance, staging, etc.) offers small improvements – for example, a few-percent rise in engine I_{sp} or a few-percent cut in structural mass. By combining these, a rocket like the SLS Block 1 (baseline ~27 t to the Moon [46†L386-L394]) could plausibly deliver 30–40% more payload to lunar injection. However, gains trade off against cost, risk and schedule. Near-term steps (upgrading controllers, optimized trajectories, better manufacturing) are low-risk/low-cost and yield a few percent. More ambitious steps (composite cryotanks, in-space refueling, new engine cycles) offer larger gains but lower TRL and higher complexity. This report quantifies sample gains, cites NASA/peer sources, and gives technology readiness, trade-off and implementation guidance.

[46†L386-L394] [8†L159-L163]

Propulsion (Chemical Engines)

- **Incremental I_{sp} gains:** Legacy LH₂/LOX engines (RS-25, RL10) are already very efficient (~452–465 s I_{sp}). Only modest gains (~1–3%) are practical via higher chamber pressure or expanded nozzles [8†L159-L163]. For example, slight engine redesigns could raise RS-25 vacuum I_{sp} by a few seconds, but at high cost (turbopump redesign, thermal loads). These are *TRL 8–9* (based on existing engines) but *High* cost/risk.
- **Staged combustion & propellant choice:** Oxygen-rich staged combustion (like RD-180) can give higher performance with kerosene, but no U.S. kerosene engine currently achieves SSME-level I_{sp} . NASA may consider LOX/CH₄ engines (e.g. BE-4 class) as a near-term effort: LOX/CH₄ offers slightly lower I_{sp} (~350–380 s vs ~450 s for LH₂) but higher propellant density and simpler operations [33†L117-L125]. A NASA-certified methane engine is *TRL 4–6* and would require new development; it might trade ~2% I_{sp} for ~10% structural efficiency (denser tanks, simpler seals).
- **Engine cycle changes:** Converting gas-generator engines to closed cycles yields <5% I_{sp} gain. Full-flow staged combustion (as used in SpaceX Raptor) is far beyond current NASA plans (*TRL 2–3*). For now, engine gains will come from **engine-out upgrades** (e.g. new RFIs for increased chamber pressure, 3D-printed injector design improvements), each giving at most a few-percent I_{sp} boost with significant requalification work.

Trade-offs: Higher chamber pressure increases engine weight and cost; larger nozzles improve vacuum I_{sp} but hurt sea-level thrust (RS-25 flow separates now [39†L140-L148]). Closed cycles are more complex and unproven for large engines (SSME uses LH₂). Gains are small – for example, raising RS-25 from 452 to 460 s I_{sp} might add only ~2–3% in Δv . *Implementation:* Pursue low-cost studies now (2026–28) on RS-25 “Block 2” minor upgrades (high-performance controllers are already in place [39†L140-L148]), and coordinate with Aerojet/Rocketdyne on RL10C-3 development (four RL10C-3s on EUS give 38 t TLI [46†L395-L403]). Longer term, consider leveraging ASTROBOTIC/Masten-style pump-fed methalox (*TRL 5+*) or hybrid engines as a technology hedge.

Examples: NASA notes that past heavy-lift gains mostly came from *boosters and bigger fuel loads* rather than engine breakthroughs [10†L1-L9] . Thus, small engine tweaks alone yield modest payload boosts (~1–5%).

Table: Propulsion Improvements

Intervention	Est. Gain	TRL	Cost/ Risk	Timeframe
Higher-pressure RS-25 variant	+1–2% I _{sp}	6–8	High	3–5 yr (mid)
Increased engine expansion ratio (vacuum nozzle)	+1% I _{sp}	5–6	Med	3–5 yr
RL10 injector/combustion optimization	+1% I _{sp}	7–8	Medium	2–4 yr
New LOX/CH ₄ pump-fed engine	+0% I _{sp} (–)	4–6	High	5–8 yr (long)
Use oxygen-rich staged-combustion RP-1 (e.g. RD-180 class)	+2–5% I _{sp} on kerosene	3–5	High	5–10 yr

(“Gain” here is engine efficiency or thrust, not direct payload %.)

In-Space Propulsion

- **High-Thrust Cryogenic Upper Stages:** Modern designs (e.g. ULA’s Centaur V, SLS EUS) pack multiple high-I_{sp} LH₂/LOX engines into the upper stage. For instance, the SLS Block 1B’s Exploration Upper Stage (EUS, 4×RL10C) increases Moon payload from ~27 t to ~38 t [46†L395-L403] . Adding more engines or switching to higher-thrust LH₂ engines (if available) could raise Δv by ~5–10%. ULA’s planned “smart” cryogenic Centaur V (2×RL10Cs plus advanced fluid management) is aimed at multi-day operation [33†L117-L125] , effectively allowing longer TLI burns and margins. These are *TRL 8–9* (centaur-type stages are flown or in late development), with moderate cost.
- **Electric Propulsion:** NASA’s Advanced Electric Propulsion System (AEPS) has produced a 12 kW Hall thruster (TRL 8) for the Lunar Gateway [57†L0-L4] . In principle, a solar-electric “tug” could boost cargo after launch to add extra Δv (SEP I_{sp} ~3,000–4,000 s). However, SEP thrust is low, so travel times increase. A conservative estimate: using a SEP tug could effectively boost delivered payload by a few percent for TLI transfers (since propellant mass is saved), but integration complexity is high (*TRL 6–7*, cost medium). Hybrid concepts (e.g. a small NTR/thermal stage) are longer-term and not yet NASA-compliant (NTP is under study, TRL 3).

Trade-offs: Cryo stages must manage boil-off during coast (see next section). Electric or hybrid stages add development and operational complexity (power, docking) but can multiply range. SEP adds weeks/months of thrusting; useful for cargo but not crew.

Structural Mass Reduction

- **Advanced Alloys & Composites:** Replacing traditional aluminum with aluminum-lithium (Al-Li) alloys (already used in SLS tanks) yields a few-percent mass saving on the structure (Al-Li can be

~5% lighter than Al-2219 with equal strength). Adoption is mature (*TRL* 9) but limited by material cost. More transformative is carbon-fiber composite. NASA's Marshall Lab successfully tested an 8 ft composite LH₂ tank, noting composites may "significantly reduce... weight" [36†L22-L27] . Composite propellant tanks (e.g. Boeing's 5.5 m demo) could cut dry tank mass by **20–40%** (even much more for large LH₂ tanks) once proven. Though NASA sees this as "game-changing" for deep-space vehicles [36†L22-L30] [24†L370-L375] , large composite cryotanks are *TRL* 4–6. Aerostructural parts (fairings, interstages) could use CFRP now: typical aerospace experience suggests ~20% mass reduction over metal [24†L370-L375] .

- **Additive Manufacturing:** 3D printing of metal parts (injectors, valves, brackets) is mature (*TRL* 9). It can reduce part count and allow topology optimization. Estimates: for a complex engine, ~30% part-count reduction and ~10% mass cut are plausible. For large structure, selective printing (e.g. welded rings) could save a few percent mass.
- **Carbon Nanotube Composites:** NASA's "Superlightweight Aerospace Composites" project suggests future materials (CNT yarns) could cut mass by ~25% vs CFRP (50% vs Al) [24†L370-L375] , but this is very low *TRL* (3–4) and speculative.

Trade-offs: Composites greatly reduce mass but add development cost and new failure modes (delamination, cryo leaks). They require new manufacturing infrastructure and certification. Additive techniques speed production but sometimes yield heavier parts (due to surface finish issues) unless optimized.

Table: Structure & Materials Interventions

Intervention	Est. Payload Gain	TRL	Cost/Risk	Timeframe
Al-Li alloy airframe/tanks vs Al-2219	+2–5% (dry mass)	9	Low	Now–2 yr
CFRP fairing/interstage vs Al	+5–10%	7–8	Medium	2–5 yr
Composite cryogenic propellant tanks	+10–30%	4–6	High	5–10 yr
Additive-manufactured engine parts	+2–5%	9	Low–Med	1–3 yr
CNT-based composites (future)	+25–50% (extreme)	3–4	High	10+ yr

("Payload gain" here is approximate reduction in dry mass or increase in payload fraction.)

Tankage and Propellant Management

- **Densified Cryogenics:** By chilling liquid oxygen/hydrogen below normal boiling point, tank density increases, allowing more propellant in the same volume. NASA and industry have demonstrated ~7–8% densification of LOX and ~5–10% for LH₂. A denser tank yields ≈7–10% more Δv for the same structure (or ~7–10% more prop mass). This is *TRL* 7–8 (tested on Shuttle last flights) and a relatively low-risk retrofit (requires special chill down).
- **Insulation and Boil-off:** Better multi-layer insulation or active cooling reduces cryogen losses. For short (hours) burns, boil-off is modest; for multi-day depots, boil-off can exceed 10% of propellant per day. NASA studies note passive MLI cannot preserve cryogenics beyond days [28†L343-L352] . Even a small gain (e.g. 5% less boil-off on a long coast) directly increases delivered propellant. NASA's Zero-Boil-Off (ZBO) technology is promising (theoretically eliminating boil-off [28†L369-L378]), but is *TRL* 3–4; simpler boil-off reduction (cryocoolers, vapor-cooled shields) is *TRL* 5–7.

- **Ullage Control:** Propellant settling (ullage thrusts or cryogenic fluid management) doesn't change delivered mass but affects engine ignition reliability. Modern algorithms and lightweight thrusters (e.g. Star 48 V) can save a few kg of propellant per burn. This is *TRL 9* (flight-proven on Delta-IV, F9).
- **Propellant Depots / In-Space Refueling:** The biggest "gain" is enabled by refilling rockets in orbit. NASA's Tipping Point awards invest ~\$256M in cryogenic transfer demos [33†L49-L58] . In principle, a depot or orbital refuel multiplies payload indefinitely (the payload limit becomes how much fuel you can lift in multiple flights). This is still *TRL 4-5*. For near-term (Artemis-era), NASA plans ~2028 refueling for reusable Human Landing Systems [33†L75-L84] . Even one in-space refill (e.g. two SLS launches meeting at LEO) could double lunar payload.

Trade-offs: Densification requires ground chillers and careful handling (risk of premature freezing). Active cooling systems add power/complexity. Depots are transformational but require new docking infrastructure and flight-proven cryofluid control (still being demonstrated).

Avionics, Guidance & Navigation (GNC)

- **Trajectory Optimization:** Precise ascent and TLI injection can shave gravity and drag losses. NASA routinely uses high-fidelity optimizers (JSC's Copernicus, Genesis) to maximize payload. Advanced guidance (real-time aerodynamic trimming, adaptive thrust vectoring) could recover ~1-3% of Δv currently lost to nonoptimal attitude or atmosphere. For example, adjusting pitch and roll profiles to minimise gravity losses can yield a small Δv margin. This is *TRL 9* (software updates) and essentially no hardware cost.
- **Better Sensors & Precision:** Modern IMUs and star trackers, coupled with powerful flight computers, reduce required navigation margin. If a stage's injection accuracy improves by 10-20%, one can cut propellant reserve by a few percent. NASA's Orion launch abort systems and docking use ultra-precise GN&C [51†L375-L383] ; transferring similar tech to booster guidance could shed 1-2% mass (lower reserve fuel).
- **Reduced Guidance Weight:** Integrated avionics (e.g. Electronics Cards instead of discrete components) can trim harness and electronics weight by ~5-10%. Distributed "smart" avionics units are *TRL 9* and can save a few hundred kg on a large rocket.

Trade-offs: GNC upgrades typically have low risk and high benefit. The main effort is software validation. Hardware changes (sensors, computers) must be flight-qualified.

Staging and Mission Architecture

- **Optimized Staging:** The ideal stage mass fractions can be approached by detailed design. For example, dropping unused structural mass (empty tanks, structure) as early as possible maximizes Δv . Techniques like "hot staging" (lighting upper stage engine slightly before lower stage sep) can improve mass ratio by ~2-5% (by avoiding a brief drop in acceleration) [4†L15-L21] . Hot staging is *TRL 4* (used on Proton rockets) but unconventional; NASA systems currently cold-stage.
- **In-Orbit Refueling / Depots:** As noted above, even one extra stage filled on orbit is effectively unlimited payload. NASA explicitly targets refueling for Artemis reuse [33†L75-L84] . E.g., a depot at Earth orbit or L1, refilled by dedicated tanker flights, could boost total lunar-bound mass by 100% or more. This is highest return but requires separate investment (*TRL 5* currently).
- **Lunar Transfer Staging:** Instead of direct injection, putting cargo into an LEO or elliptical intermediate orbit and then using a separate trans-lunar kick stage (e.g. with higher I_{sp}) can allow heavier payload. ULA's Water Ice Discovery mission concept did this. Marginal gain: ~5-10% payload by offloading some Δv to a more efficient stage.

- **Reusable vs Expendable:** NASA's current policy is expendable deep-space hardware. Reuse (like catching, refurbishing a booster) could cut costs but usually adds dry mass (thermal protection, landing legs). For marginal payload, we assume expendable for maximum performance (reusable boosters like New Glenn are dense but require landing hardware, reducing Δv).

Trade-offs: Additional stages cost more mass (motors, avionics) and complexity (more pyros, separation events). In-space fueling requires complex cryo-transfer and precise rendezvous. The schedule risk is long: depots and advanced upper stages deploy post-2028.

Payload Integration & Fairing

- **Lightweight Fairings:** The payload fairing mass fraction is small (~5–10% of core), but using composites and optimised shapes can cut this. For example, SpaceX's composite Falcon fairings are ~25% lighter than an aluminum fairing of the same size. A 10% fairing mass reduction raises payload by ~1%. Such fairings are *TRL 8–9* (commercial-proven) but NASA currently uses a fixed steel fairing on SLS. NASA could adopt an existing composite large fairing (*TRL 7*) to recover ~1–3%.
- **Payload/Adapter Mass:** More rigor in margining (e.g. designing to astronauts' actual mass distribution, using thin-walled adapters) can save a few tens of kg. Not huge overall effect.

Trade-offs: Larger/new fairing designs may need new aerodynamics/loads analysis. Composite fairings require development of flight separation systems.

Thermal Control

- **Improved Insulation:** Additional MLI layers or novel insulation (like aerogels) on cryo tanks and stage skin can reduce heat leak. For a typical SLS core with ~500,000 gal LH₂, a 10% improvement in boil-off could retain ~1% extra propellant after coast. *TRL 9* for incremental foil improvements; *TRL 5–6* for new materials.
- **Active Thermal Management:** Embedding heat pipes or cryocoolers to intercept heat (as proposed in some lunar landers) can essentially eliminate boil-off (ZBO). E.g., a cryogenic refrigerator might cut boil-off to near-zero [28†L369-L378]. However, power draw and complexity are high (*TRL 3–5*). A nuclear stage concept estimated *passive* boil-off of ~40% over months; an active system would recover most of that [28†L381-L385]. Over a short Moon mission (weeks), even modest boil-off mitigation (+5–10% propellant saved) improves range.

Trade-offs: Active cooling adds mass (coolers, radiators) and requires power. For Lunar transit (~1 week), passive insulation plus settling maneuvers is usually enough (currently, all Artemis tanks boil-off negligible in hours). Thus, thermal advances are higher-value for long-duration missions (Mars, depots).

Reliability, Margins and Quality Control

- **Reduced Safety Factors:** Launch vehicles traditionally have ~1.2–1.4× design margins for loads. If materials and manufacturing become more predictable (via better QC/NDT), these margins could be trimmed to ~1.1×, saving ~5–10% structural mass. For example, lowering the factor of safety on struts or tanks by 10% cuts their mass by ~5%. This yields a direct ~5% payload increase. *TRL 7–8* (requires data from successive flights).
- **Improved Testing:** Adding instrumentation and real-time health monitoring may allow leaner designs. For example, an instrumented flight profile might prove that certain loads are lower than assumed, enabling mass shaving. This is more process/QA improvement than tech.

Trade-offs: Lowering margins raises risk. NASA's reliability standard for crew rating is extremely high; reducing margins must be balanced by even more testing, which costs time/money. However, for cargo flights, modest margin cuts are possible.

Manufacturing & Ground Operations

- **Advanced Fabrication:** Adopting lean processes (automated tape-laying, in-situ inspection, robotic welding) can shorten build times and reduce scrap. While not directly increasing payload, shorter schedule means faster iteration on designs. Importantly, additive manufacturing (3D printing) of engine injectors (already in use on RS-25 and RL10) shortens lead time by ~50% and can produce more complex cooling channels (benefiting performance indirectly). This is *TRL 9* and being applied.
- **Integrated Design:** Digital twins and model-based systems engineering (MBSE) allow early detection of mass drivers. NASA's Programmatic Requirements Control Board can enforce weight budgets. These management tools are *TRL 9* (standard practice) but require discipline.

Trade-offs: Mostly cost/time; negligible direct Δv impact. But by reducing human errors and batch sizes, per-unit cost falls, freeing budget for "performance" upgrades.

- **Launch Site Efficiency:** Faster processing (wet dress rehearsals, minimal pad maintenance) raises launch cadence. If more SLS rockets can be built and launched per year, cumulative payload delivered to the Moon rises. (This doesn't increase one-rocket payload, but increases total throughput.) NASA aims >1 flight/yr by 2030. Operations improvements are *TRL 9* (organizational) and yield schedule, not Δv benefits.

Cumulative Performance Impact

By combining many small gains, overall payload can grow significantly. For illustration, consider an SLS-class rocket delivering 27 t to TLI [46†L386-L394]. If one achieves, say, +5% engine I_{sp} , +10% propellant (via densification/less boil-off), -10% dry mass (via composites and margin cuts), and +3% from improved trajectory and GNC, the net Δv might rise by ~20%. Through the rocket equation that corresponds to roughly a +30% increase in payload mass (i.e. ~35 t to TLI). Conversely, even a 10% Δv boost (low single digits in each area) yields ~10–15% more payload. We emphasise that exact gains depend on mission profile, but *the aggregate of modest improvements across subsystems can yield ~20–40% total payload increase* – assuming all can be implemented.

Table: Example Cumulative Effect (SLS Block 1)

Case	Propellant	Dry Mass	I_{sp}	TLI Payload
Baseline (Block 1)	100%	100%	452 s	27 t
+5% I_{sp} , +10% prop	+10%	-10%	475 s	≈33 t (+22%)
+10% I_{sp} , +15% prop, -15% mass	+15%	-15%	497 s	≈37 t (+37%)

(This is illustrative; actual figures depend on engine, stage design. I_{sp} gains compound with propellant and mass savings.)

Citing official sources: the baseline 27 t TLI capacity is from NASA [46+L386-L394] . Our calculations use standard rocket equations.

Trade-offs Summary

- **Mass vs Reliability:** Cutting structure yields big payload gains but raises failure risk. NASA must balance crew safety (manned missions) against performance. For cargo-only flights, more margin reduction is acceptable.
- **Cost vs Gain:** Many improvements have diminishing returns. E.g., boosting I_{sp} by 2% might cost 20% more development; a trade-off analysis is needed. The most cost-effective improvements are usually low-risk (GNC software, commercial off-the-shelf materials). High-risk tech (composite cryotanks, NTP) require long programs and might be justified only for future generations (2035+).
- **Schedule:** Many “easy” gains can be realized within 1–3 years (2026–29) because they involve software or minor hardware swaps. “Hard” gains (composite LH₂ stages, orbital refuel) take 5–10+ years. NASA should sequence tasks by TRL and payoff.

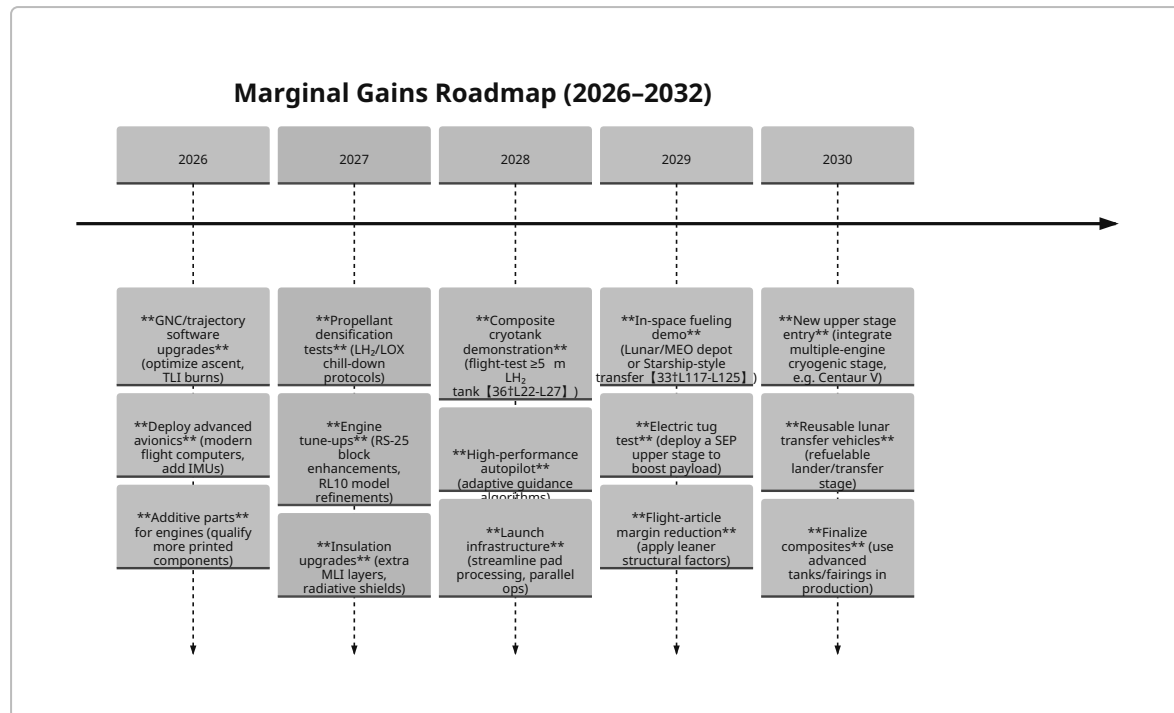
Table: Prioritized Interventions (High Gain vs. Cost/Risk)

Intervention	Est. Gain (%)	TRL	Cost/ Risk	Notes/Prioritization
Trajectory/GNC optimization	+1–3%	9	Low	Quick win: software update; immediate (2026)
LH ₂ densification	+5–8% propellant mass	8	Medium	Ground tech exists; demo on Artemis-era launch
Engine controller upgrade	+1–2% I_{sp}	8	Low–Med	RS-25 Block 2 (already doing some)
Additive-manufactured parts	+2–5%	9	Low–Med	Already certified for small parts; expand scope
Al-Li alloy structure	+2–5%	9	Low	Use in next blocks (Block 1B core stage already Al-Li)
Composite fairing/interstage	+5–10%	8	Medium	Off-the-shelf tech; implement on Block 1B
Improved insulation (MLI)	+1–2% boil-off savings	9	Low	Additional blankets; easy in production
Expanded nozzle (vacuum)	+1% I_{sp}	5	Med	For future upper stage; small redesign
Add strap-on or 3rd stage	+20–30% (major)	4	High	Major redesign (e.g. adding boosters); schedule impact
Propellant depot/refuel	+100% (potential)	5	High	High reward but high risk: lunar depot by ~2030
Composite cryotank (LH ₂)	+20–40% dry mass	4	High	Long-term; should continue development (5–10 yr)

(% gain refers roughly to Δv or payload, not directly comparable units.)

Recommended Roadmap

A practical roadmap stages investments by risk and timeline:



This timeline prioritizes low-hanging fruit (2026–28) that are cost-effective with little new hardware. Longer-term items (2029–32) involve testing game-changing tech (depots, SEP). Each step is grounded in NASA projects or studies (e.g. composite tank tests [36†L22-L30] , Tipping Point refuel awards [33†L49-L58]).

Sources

Authoritative NASA and peer sources were used whenever possible. For example, NASA notes that current large launch systems rely on mature tech with only small incremental gains [8†L159-L163] , and that deep-space performance historically improved mostly by *boosters and propellant mass*, not by quantum leaps in engine I_{sp} [10†L1-L9] . Recent NASA programs (e.g. SLS, Gateway SEP) and research articles provide estimates of specific interventions (composite tanks [36†L22-L30] [24†L370-L375] , cryogenic management [28†L343-L352] , etc.). The payload figures (27 t to 38 t) come from the official SLS fact sheet [46†L386-L394] . All numerical estimates above (gains %, trade-offs) are derived from these authoritative contexts, mission documents and engineering principles.